STILLSTACK · PRODUCT REQUIREMENTS AND SPECIFICATIONS

StillStack product requirements and specifications

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DOCUMENT CONTROL

Document ID	SSK-PRS-001	Version	0.10
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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stillstack		
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REVISION HISTORY

Version	Date	Author	Change
0.10	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 ("i concur with your recommendations"): PR-002 restated as 1.2 or more for four stages (D-10), not met to met on paper; PR-013 stated for the anchored stand with anchoring a hold point and the cover rule on the panel label (D-09, D-05), not met to met on paper; PR-010 clips 5052 aluminium (D-08); PR-012 USD 460.70 with the panel label
0.9	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: REQ migrated to PRS (SSK-PRS-001; was SSK-REQ-001); R1 to R12 become PR-001 to PR-012 with the same targets, verification and status, traced to SSK-URS-001; PR-013 (wind) added from the CAL; PR-008 and PR-012 updated for the stage 3 PPS rails (SSK-DEC-001 D-01); status from SSK-CAL-001 v0.8 after the independent review (PR-001 at risk, PR-002 not met, PR-013 not met unanchored)

0.9	2026-10-02	Amish Chadha	Approved follow-ups carried into the design: R9 mass 18.7 kg, R12 USD 443.20 (USD 88.20 over the target), R4 lip modelled, R10 moves from at risk to met on paper (wick clips modelled)
0.7	2026-10-02	Amish Chadha	Decisions of 2026-10-02 noted under R4, R6, R7, R8 and R10 (lip, conductivity pass condition, fabric test, coating rule, PPS, wick clips); no status changed
0.6	2026-10-01	Amish Chadha	Status from the constructable design (SSK-DDR-003); R12 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Add TRL 3 status from SSK-CAL-001; record decisions from SSK-DDR-001 (budget kept at $250, tilt stand, aperture); targets unchanged
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

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What the design must do, each requirement traced to a user need in SSK-URS-001. Targets are proposals for review with the co-design partner. Method: STANDARDS section 22. This document replaces SSK-REQ-001 (kit 1.8); each PR keeps the target, verification and status of the R it replaces, with the old number in "Was". Status comes from SSK-CAL-001 v0.7 and the STEM check SSK-STM-001.

No requirement is **not met**. PR-002 (gained output ratio) and PR-013 (wind) were not met after the independent review of the STEM check; Amish decided on 2026-10-25 ("i concur with your recommendations") to restate PR-002 as 1.2 or more for four stages (D-10) and PR-013 for the anchored stand, with anchoring a hold point (D-09), so both are now met on paper. One is **over target**: PR-012, parts cost USD 460.70 against the USD 355 value-engineering target (USD 105.70 over; Amish set budgets as value-engineering targets, not limits, on 2026-10-01). Five are **at risk** (PR-001, PR-004, PR-006, PR-007, PR-008) and one can only be verified at TRL 4 (PR-003). The underside and fins are now painted black (SSK-DEC-001 D-07), without which PR-001 would not be met. The stage 3 rails moved from printed polycarbonate to PPS on 2026-10-24 under Amish's standing instruction of 2026-10-23 (SSK-DEC-001 D-01, STEM row S11), which gives PR-008 a real margin in stage 3 and adds USD 16 to PR-012.

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001, UR-002	Performance	Daily distillate	The panel shall produce at least the target distillate per m2 of aperture on a clear 5.5 kWh/m2 day (design goal 13 L)	10	min	L/m2/day	Analysis (SSK-CAL-001 section 4), TRL 3; outdoor test beside a basin still at TRL 4	At risk: 10.3 L after warm-up (10.9 quasi-steady), 3 % margin, with the painted underside of D-07; 10.1 L quasi-steady at a glazing transmittance of 0.75; 13 L goal not met	R1
PR-002	UR-002	Performance	Heat reuse	The daily gained output ratio (latent heat in the distillate over solar input) shall reach the target for four stages (restated from 1.3 by Amish, 2026-10-25, D-10)	1.2	min	ratio	Analysis (SSK-CAL-001 section 4), TRL 3	Met on paper: 1.29 quasi-steady, 1.22 after warm-up (1.7 % margin); 1.20 at a glazing transmittance of 0.75	R2
PR-003	UR-003	Safety	Distillate salinity	Distillate conductivity shall not exceed the target (about 50 mg/L dissolved solids) for feed up to 40 g/L	75	max	µS/cm	Analysis of carryover paths (SSK-CAL-001 section 7), TRL 3; conductivity meter per stage at TRL 4 (pass condition before any water is drunk)	To verify at TRL 4: brine carryover must stay below 8.2 mL/day (0.075 %)	R3
PR-004	UR-004	Safety	Brine separation	Every wick edge shall keep a physical air	10	min	mm	Inspection of the model (`model.py`	At risk: 10 mm breaks at ribs, rails and plate edges,	R4

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				break from every distillate path, with no shared drain, and the outlets shall be labelled DISTILLATE and BRINE				`--check`), TRL 3; conductivity per stage at TRL 4	brine tails 10.5 mm over the closed lid and 44 mm from any distillate tongue, lip tabs on the high edges; carryover proven only by test	
PR-005	UR-005	Performance	Passive operation	The panel shall run with no pump, electronics or battery, fed by gravity from a container no higher than the target above ground	1.5	max	m	Inspection of the design, TRL 3	Met: passive; trough rim at most 1.00 m (at 35 deg)	R5
PR-006	UR-006	Performance	Feed water	The panel shall accept feed up to 40 g/L after a 20 µm or cloth prefilter at a feed-to-distillate ratio of at least the target, and the wick shall carry that feed	2.5	min	ratio	Analysis (SSK-CAL-001 sections 6 and 7), TRL 3; wicking-rise test of the shortlisted fabrics	At risk: brine 66.7 g/L, far below saturation; wick permeability of 2.9e-11 m2 needed (3.0e-11 used for the test), not yet measured	R6
PR-007	UR-007	Materials	Food-contact wetted parts	Every surface that touches distillate shall be food-contact rated (316 stainless, certified	100	min	% of wetted area	Review of the material list and certificates, TRL 3	At risk: fluoropolymer coating class chosen; no named product with a drinking-water certificate	R7

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				coated aluminium, PP, HDPE or NSF/ANSI 51 or 61 silicone)					seen (rule of 2026-10-02: else 316 stainless for plates 1 and 2)	
PR-008	UR-008	Environment and durability	Dry stagnation	After an empty-wick day at 1,000 W/m2 and 35 °C, calm or windy, there shall be no damage: stages 1 and 2 rated to 150 °C or more, every other internal material to 110 °C or more, a stagnation cover supplied and a shade rule written	150	min (stages 1, 2); 110 elsewhere	°C	Analysis (SSK-CAL-001 section 5) and datasheets, TRL 3	At risk: stages 1 and 2 reach 142 °C calm (8 K margin to the 150 °C rating, none at 1,100 W/m2; PPS rails long-term 218 °C); stage 3 reaches 107 °C calm, with PPS rails (SSK-DEC-001 D-01); inner glazing 113 °C against 120 °C (kept with the cover and shade rule as the barrier, D-05); wick and coating ratings not yet confirmed	R8
PR-009	UR-009	Physical architecture	Household scale	1.0 m2 aperture; panel dry mass no more than the target; footprint within 1.3 x 1.3 m	20	max	kg	Analysis (SSK-CAL-001 section 8) and model, TRL 3; weigh at TRL 4	Met on paper: 18.8 kg dry (22.8 kg wet); footprint 1.28 x 1.25 m at 10 deg	R9
PR-010	UR-010	Usability	Wick maintenance	One person shall remove, rinse and refit the wicks	30	max	min	Design review (build plan SSK-BLD-001),	Met on paper: plates lift out once the low wall cap is off; wicks	R10

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				without special tools within the target; salt flushing no more often than weekly				TRL 3; timed at TRL 4	held by two 5052 aluminium clips each (D-08), no adhesive	
PR-011	UR-011	Usability	Tilt range	The stand shall set the tilt from 10 deg to the target	35	max	deg	Model check at every hole (`model.py --check`), TRL 3	Met: six prop holes, 10 to 35 deg in 5 deg steps	R11
PR-012	UR-012	Physical architecture	Cost and buildability	Parts cost reported against the value-engineering target; buildable with hand tools, sheet-metal shears and a 3D printer	355	target	USD	Priced BOM (SSK-CAL-001 section 10), TRL 3	Over target: USD 460.70 (USD 105.70 over), after the stage 3 PPS rails of SSK-DEC-001 D-01 (USD 16)	R12
PR-013	UR-013	Safety	Wind and hot surfaces	Anchored by its four ground stakes or 40 kg of ballast, a hold point before the panel is filled or faces the sun, the panel and stand shall not tip or lift below the target wind speed; and the hot-surface rules (cover, shade rule) shall be in the instructions and on the	17	min	m/s	Analysis (SSK-CAL-001 section 9), TRL 3; stake pull-out in soft or sandy ground at TRL 5	Met on paper anchored: with 40 kg of ballast it tips at 28.8 m/s, and each rear stake needs about 6 N of hold-down at 17 m/s; unanchored it tips at 16.5 m/s (lift at 18.7 m/s), which the hold point excludes; cover BOM line 14, panel label BOM line 19	New

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				panel label (restated by Amish, 2026-10-25, D-09)						

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- Design day: 5.5 kWh per m2 per day (19.8 MJ per m2) on the tilted aperture, 30 °C mean daytime ambient, light wind (1 m/s in SSK-CAL-001).
- Latent heat of vaporization: SSK-CAL-001 uses temperature-dependent values (about 2.33 to 2.36 MJ per kg over the stage range).
- The WHO does not set a health-based limit for dissolved solids; 50 mg/L is a check that salt carryover is negligible, not a potability limit.
- Household need of about 3 L per person per day for drinking and 2 L for cooking, so 10 to 11 L per day serves a household of two.
- The history of each target (R numbers, SSK-DDR-001 to SSK-DDR-003, SSK-DEC-001) is in the revision table above and in the review note.